

Business Trends & Market Analysis in Los Angeles

CS163 Group 8

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SJSU CS163

INTRO

02

SJSU CS163

DATA

04

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VISUAL UNDERSTANDING

05

SJSU CS163

SURVIVAL CURVE

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SJSU CS163

ML (LSTM)

14

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ML (RSF)

17

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SUMMARIZE

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	Objective	
	The goal of this project is to assist new business owners in making smarter, data-driven decisions when starting a business by analyzing trends in the survival and closure of businesses across Los Angeles. Our tool offers customized recommendations (location district/coordinate, industry/NAICS).	
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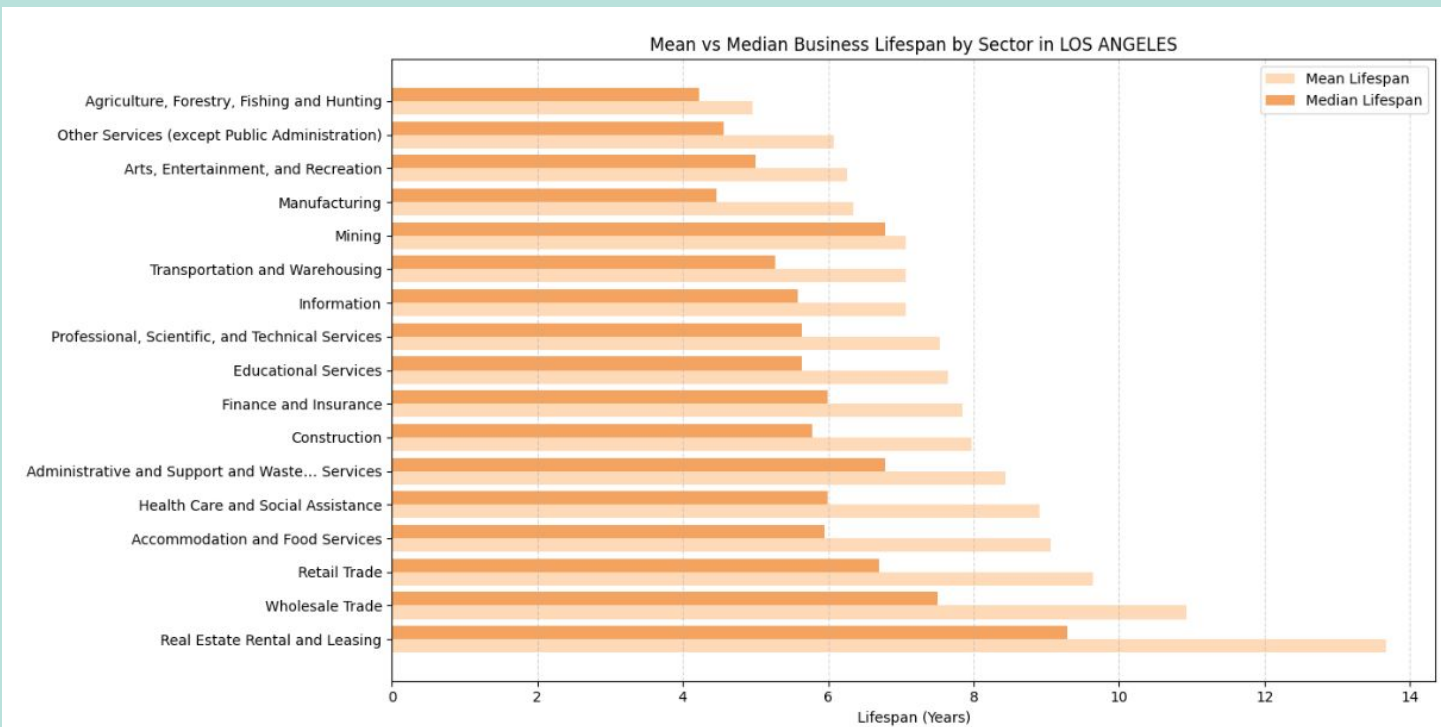
Motivation: Small businesses often fail due to misjudged location/district choices, poor financial planning, and industry risks. While extensive business data exists, it's often inaccessible or too complex for new business owners to interpret effectively. We aim to bridge that gap.

Problem Addressed:

- ❑ Lack of data-driven insights for business owners.
- ❑ Unknown failure probabilities for future decision-making in business planning.
- ❑ Limited resources to understand location-based and industry-specific risks.

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Data Sources ❏ Active and closed business records from Los Angeles city records https://data.lacity.org/Administration-Finance/Listing-of-All-Businesses/r4uk-afju/about_data ❏ NAICS codes for industry classification https://www.naics.com/search/#naics		
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Mean and Median Lifespan for Each Sector in Los Angeles



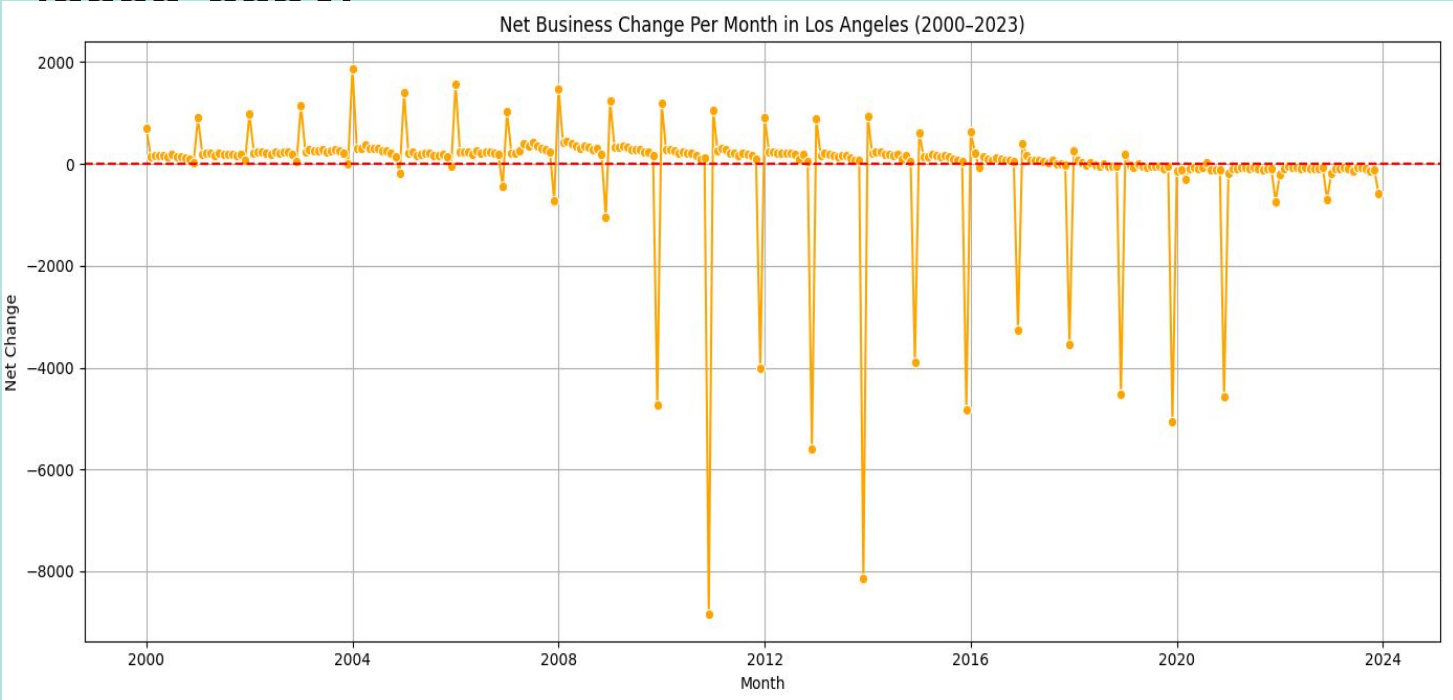
- Most industries show a right-skewed distribution.
- We can expect our ML model to encounter a large number of short-lived businesses, which could affect how predictions are made.

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	Distribution of Business Lifespan by sector in Los Angeles The violin plot displays the distribution of business lifespan across 15 sectors in Los Angeles. The sectors listed on the y-axis are: Agriculture, Forestry, Fishing and Hunting; Manufacturing; Other Services (except Public Administration); Arts, Entertainment, and Recreation; Transportation and Warehousing; Information; Educational Services; Professional, Scientific, and Technical Services; Construction; Accommodation and Food Services; Finance and Insurance; Health Care and Social Assistance; Retail Trade; Administrative and Support and Waste... Services; Wholesale Trade; and Real Estate Rental and Leasing. The x-axis represents the lifespan in years, ranging from 0 to 80. Each violin represents the density of businesses within a sector, with a thick black line indicating the median and a thinner line showing the range. Most sectors show a high density of businesses with lifespans between 0 and 20 years, with some outliers extending up to 80 years.	
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- Most businesses are clustered in terms of lifespan, while the thinner tails highlight outliers.
- It gives us a more complete picture than just averages that helping us identify central tendencies and skewness for each sector.

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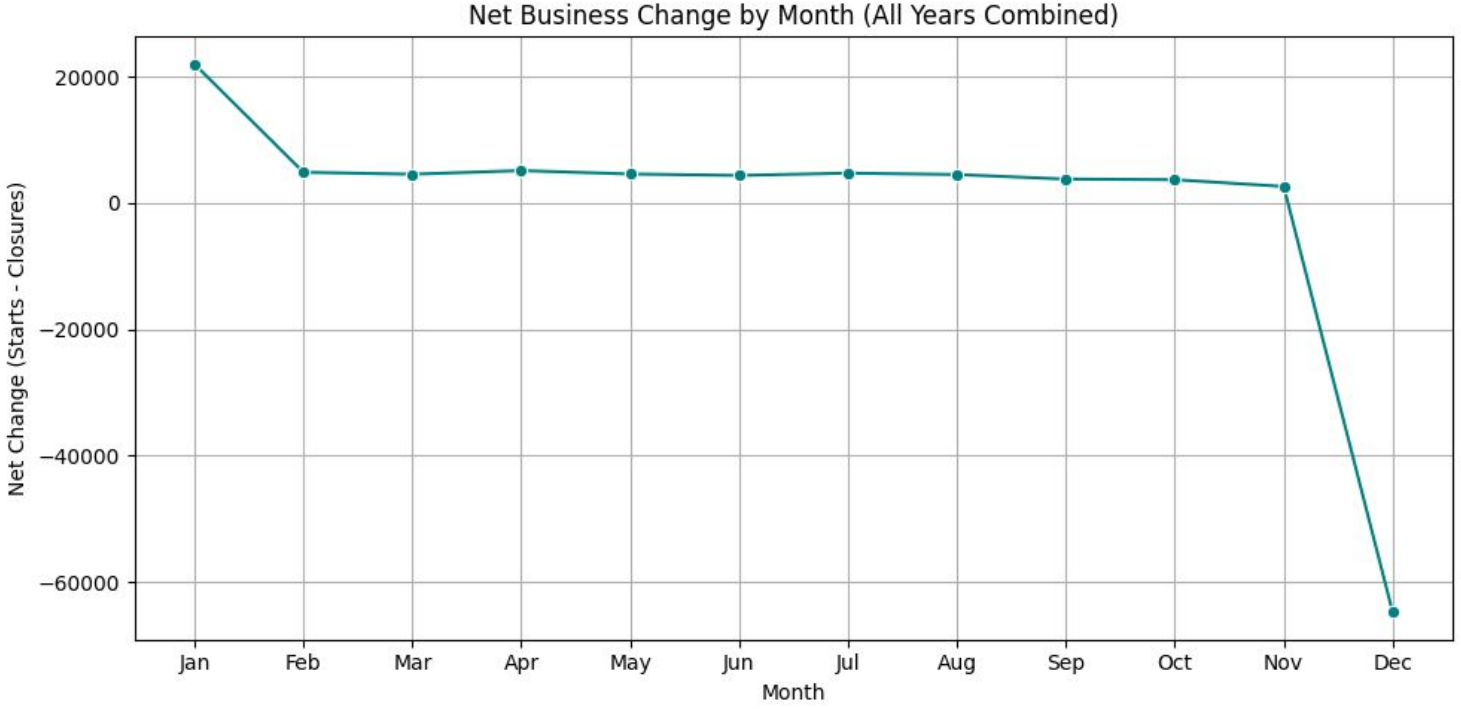
Net Business Change in Los Angeles



- This is calculated as the number of new businesses started minus those that closed each month.
- We see clear patterns of spikes and drops, including major dips during the 2008 financial crisis and COVID-19.

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Net Business Change by Month (All Years)



- Seasonal waves repeat annually, with peaks in JAN, and significantly reduced in DEC.
- This pattern highlights how macroeconomic events and policy shifts impact business dynamics over time. (Also can be proved by Kaplan-Meier)

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Business Map SELECT NAICS SECTORS APPLY Pick one or more NAICS Sectors: CLEAR ☐ 11 - Agriculture, Forestry, Fishing and Hunting (1184) ☐ 21 - Mining (95) ☐ 22 - Utilities (6) ☐ 23 - Construction (38060) ☐ 31 - Manufacturing (12233) ☐ 32 - Manufacturing (2155) ☐ 33 - Manufacturing (6310) ☐ 42 - Wholesale Trade (31694) ☒ 44 - Retail Trade (30190) ☒ 45 - Retail Trade (28703)					

Crash Course on survival analysis

Survival Probability: Probability of businesses remaining open at different time intervals.

Kaplan-Meier Method: technique to estimate survival probabilities

$$S(t) = \prod_{i:t_i \leq t} \left(1 - \frac{d_i}{n_i}\right)$$

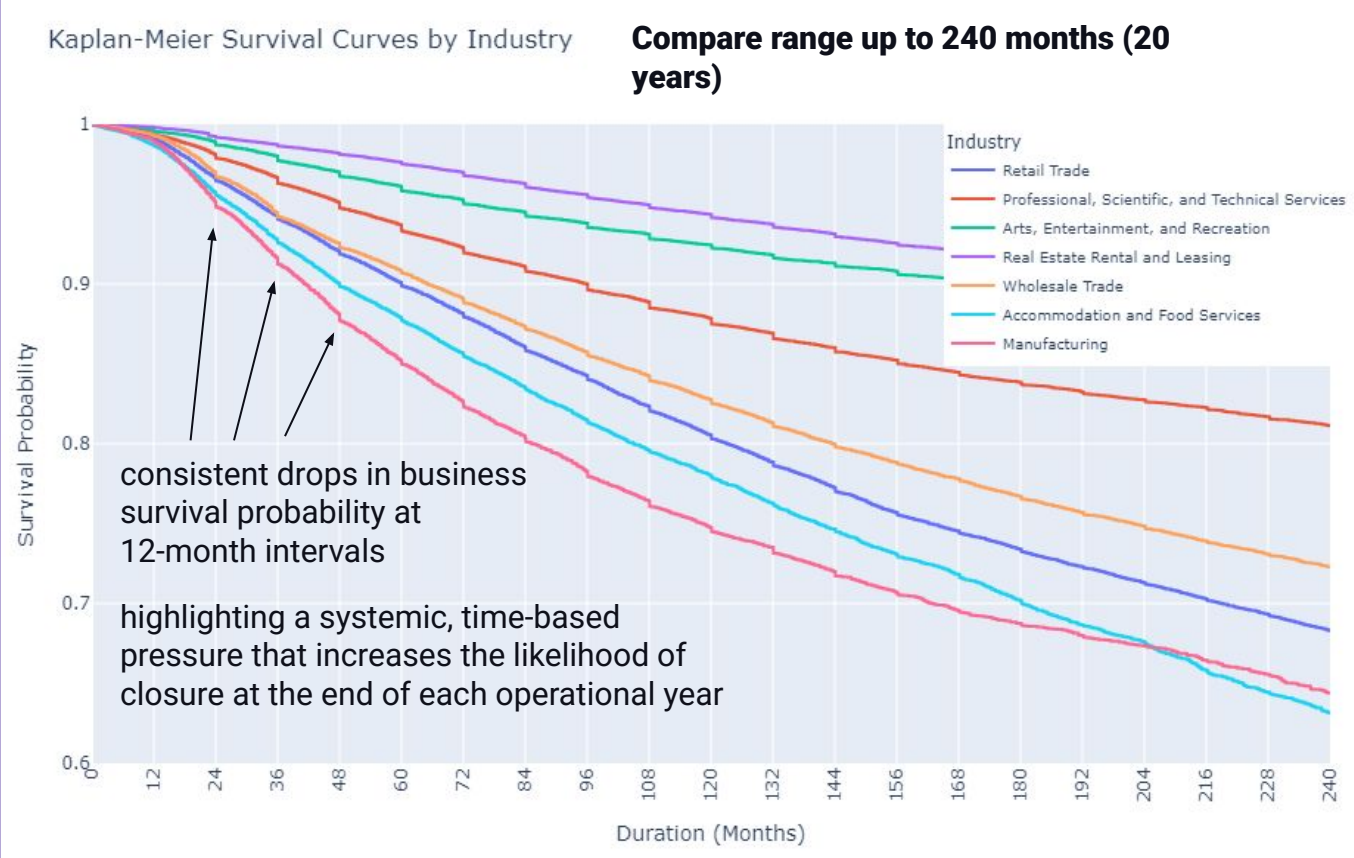
t_i = at least one event happened at time i
 d_i = number of events (business closed in our case)
 n_i = number known which survived

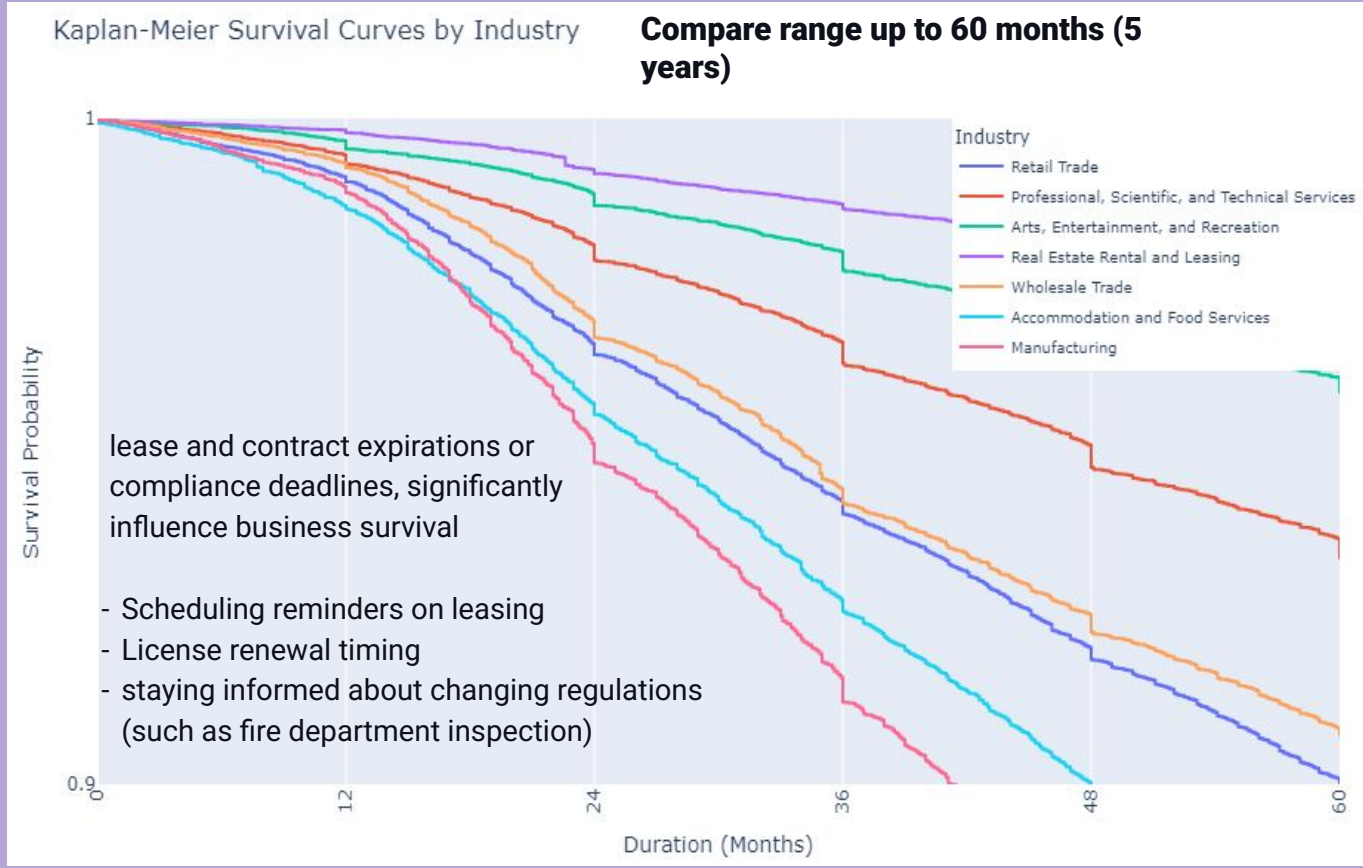
Reading Curve: X-axis: Time (months/years).

Y-axis: Survival probability (percentage of businesses still open).

Business	Start_Date	Closed_Date	Duration	Event(Close)
Business_A	2015-01-01	2019-06-05	53.0 months	1
Business_B	2016-03-25	-	108.8 months	0
Business_C	2017-10-15	2020-01-30	27.4 months	1

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Long Short Term Memory

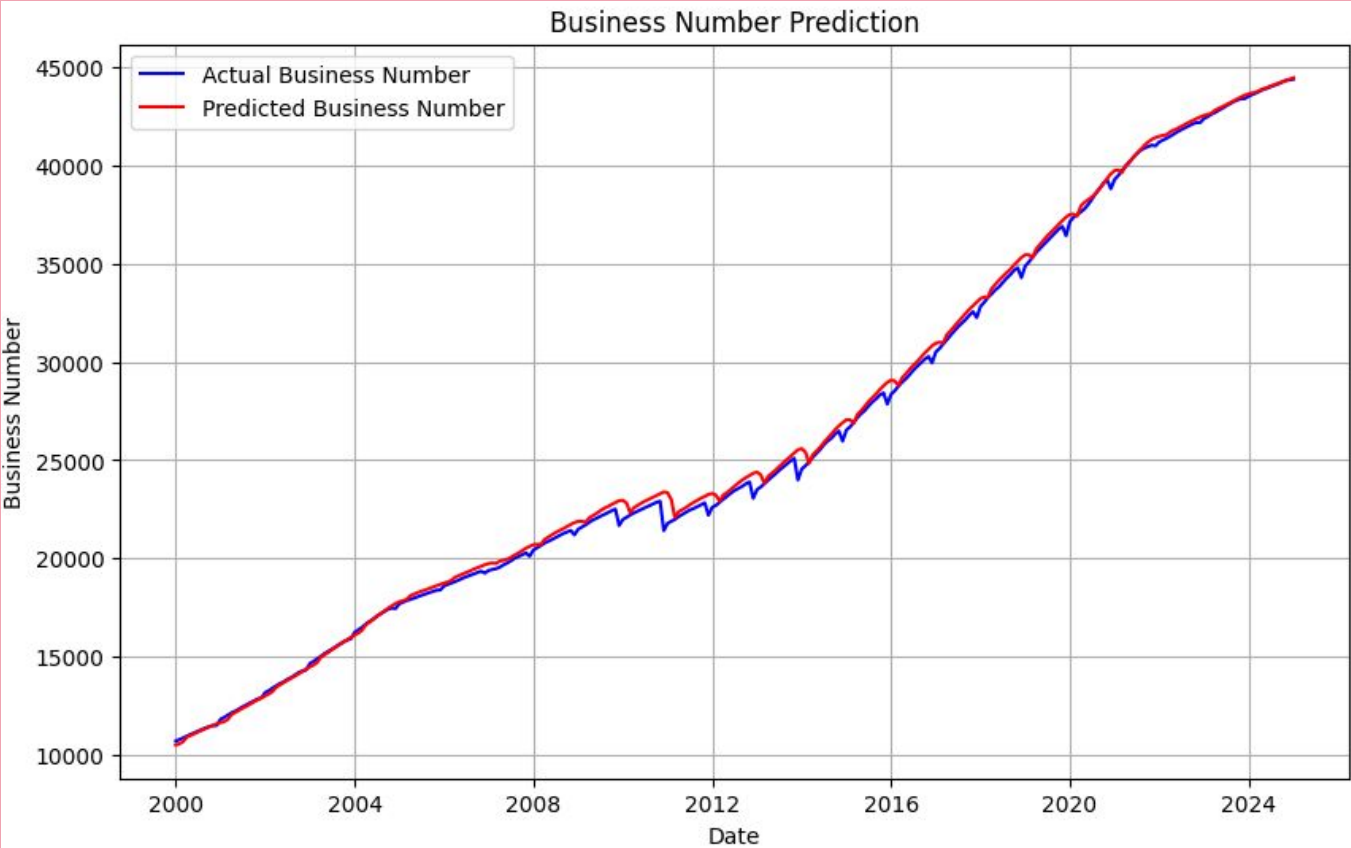
Layer (type)	Output Shape	Param #
lstm_8 (LSTM)	(None, 3, 128)	66,560
dropout_8 (Dropout)	(None, 3, 128)	0
lstm_9 (LSTM)	(None, 3, 128)	131,584
dropout_9 (Dropout)	(None, 3, 128)	0
lstm_10 (LSTM)	(None, 3, 128)	131,584
dropout_10 (Dropout)	(None, 3, 128)	0
lstm_11 (LSTM)	(None, 64)	49,408
dropout_11 (Dropout)	(None, 64)	0
dense_2 (Dense)	(None, 1)	65

Model

Sequence
& Values

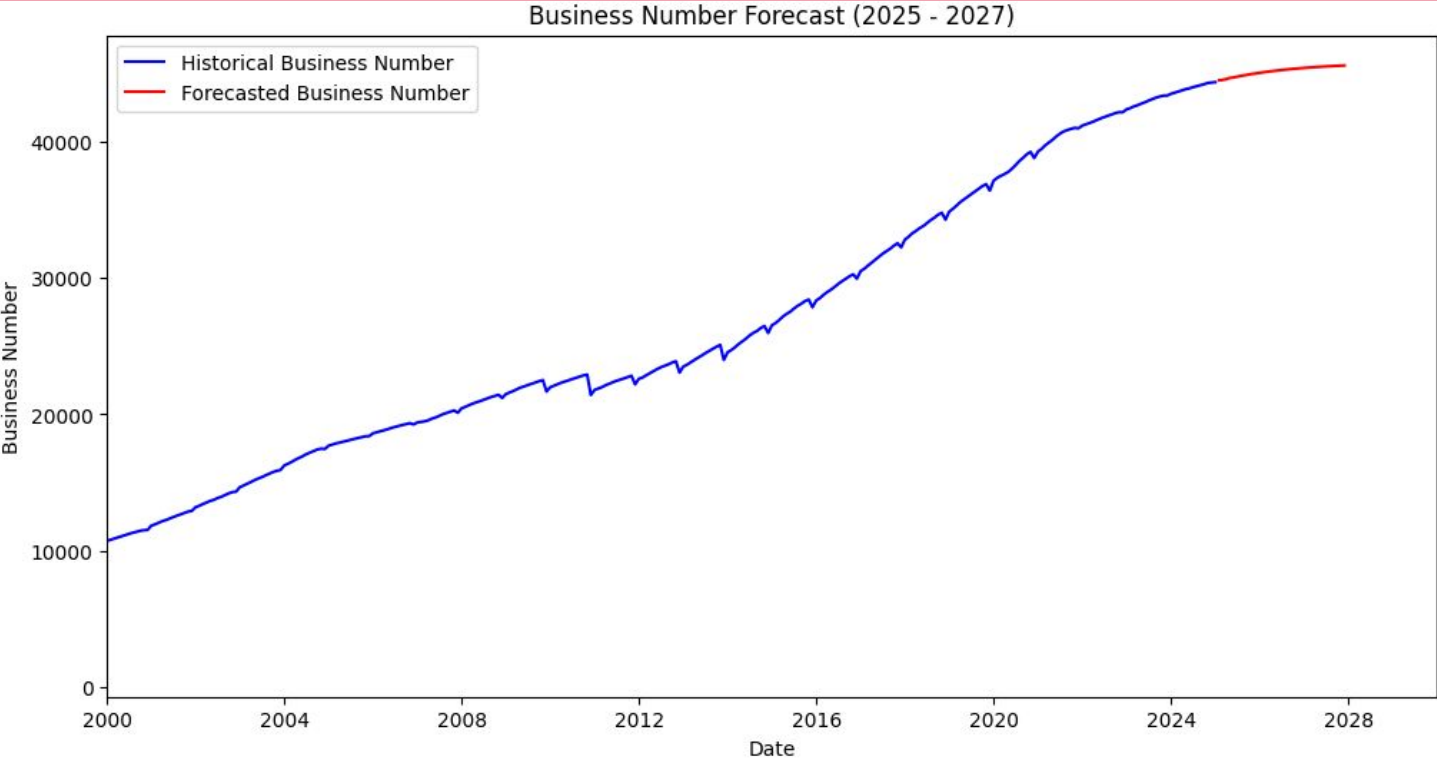
date	business_number
1921-01	1.0
1928-09	2.0
1929-01	3.0
1930-10	4.0
1931-01	5.0
...	...
2025-03	428612.0
2025-05	428615.0
2025-06	428616.0
2025-09	428615.0
2030-07	428614.0

Long Short Term Memory (Test)



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Long Short Term Memory (Prediction)



Random Survival Forest (Model)

```
RandomSurvivalForest(max_depth=5, min_samples_leaf=50, min_samples_split=100,  
n_estimators=6, n_jobs=-1, random_state=1024)
```

rsf.get_params()

```
{'bootstrap': True,  
'low_memory': False,  
'max_depth': 5,  
'max_features': 'sqrt',  
'max_leaf_nodes': None,  
'max_samples': None,  
'min_samples_leaf': 50,  
'min_samples_split': 100,  
'min_weight_fraction_leaf': 0.0,  
'n_estimators': 6,  
'n_jobs': -1,  
'oob_score': False,  
'random_state': 1024,  
'verbose': 0,  
'warm_start': False}
```

6 trees ($n_estimators=6$)

Each tree stops splitting at
depth 5 max

A node must have **at least 100 samples to split**

A leaf node must have **at least 50 samples**

Splits are based on a **random subset of features**

Model procedure of using Kaplan-Meier

Step 1: Input goes through all trees. (inputs)

Step 2: Trees split using decision rules (from last slide)

Step 3: In each leaf, build a Kaplan-Meier curve (training data)

Step 4: Repeat this for all 6 trees

Step 5: Average all 6 curves

Final Output:

Predicted survival probabilities:
 Probability of surviving at least 6 months: 49.62%
 Probability of surviving at least 12 months: 48.13%
 Probability of surviving at least 36 months: 35.34%

These are averaged from the Kaplan-Meier curves in all 6 trees.

Each one was built using 50+ training samples per leaf.

$$S(t) = \prod_{i:t_i \leq t} \left(1 - \frac{d_i}{n_i} \right)$$

t_i = at least one event happened at time i
 d_i = number of events (business closed in our case)
 n_i = number known which survived

C-index

$$\text{C-index} = \frac{\text{Number of concordant pairs}}{\text{Number of comparable pairs}}$$

Number of the model **got the order correct** for a comparable pair

Number of any two businesses we can fairly compare.

If the model predicts that **the one who lived longer** really **has a better survival prediction**, that is concordant.

Example:

Datapoints: Business A survived **40 months**, Business B survived **20 months**. **So A lived longer than B.**

Model predicted:A: **38 months**, B: **25 months**. So the model also says A survives longer, => it is **concordant pairs**.

Otherwise: discordant.

C-index value Interpretation

1.00 Perfect model

0.80–0.90 Strong ranking

0.70–0.80 Acceptable

0.50–0.70 Weak or borderline

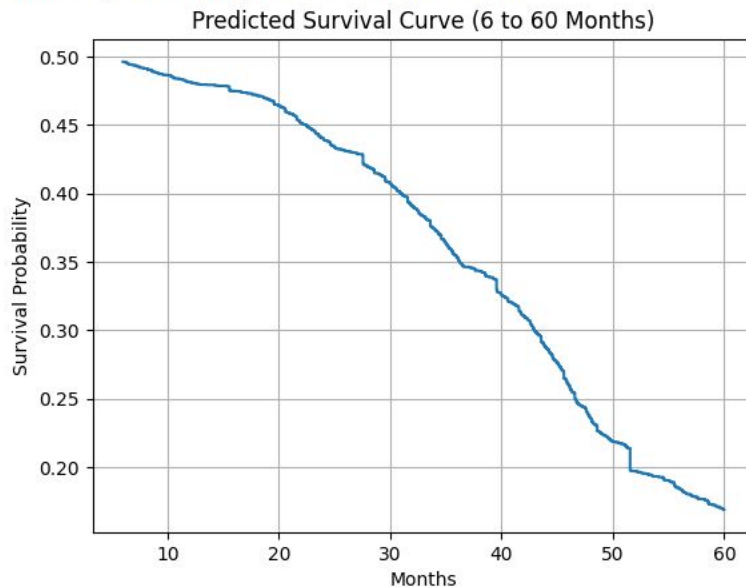
0.50 No skill (random)

< 0.50 Wrong ranking(bad)

Random Survival Forest

Concordance Index (C-index): 0.7574

```
Predict Business Survival Probability
Enter Start Year (e.g., 2026): 2026
Enter Start Month (1-12): 12
Enter Council District (0-15): 3
Enter NAICS Code (first 2 digits, e.g., 44): 53
Predicted survival probabilities:
Probability of surviving at least 6 months: 49.62%
Probability of surviving at least 12 months: 48.13%
Probability of surviving at least 36 months: 35.34%
```



Sample input and output:

Input:

Year:	2026
Month:	12
Council district:	3
NAICS code:	53 (Real Estate)

Output:

Predicted survival probabilities:

At least 6 months:	49.62%
At least 12 months:	48.13%
At least 36 months:	35.34%

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Website Link

Eric Zhao



<https://my-project-cs122-20241114.uw.r.appspot.com/>

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